Homework 08

Question 1 Answers

Part a

Since $\|(0.8, 0.6)\|_2 = 1$, the distance to the boundary is $|f(x)|$. An observation is inside the margin if $yf(x) < 1$, on it if $yf(x) = 1$, and beyond it if $yf(x) > 1$.

```
margin20 <- read.csv("data/margin-geometry-20.csv")
f <- 0.8 * margin20$x1 + 0.6 * margin20$x2
yf <- round(margin20$y * f, 10)

margin20$distance <- abs(f)
margin20$yf <- yf
margin20$position <- ifelse(yf < 1, "inside", ifelse(yf == 1, "on", "beyond"))

print(margin20, row.names = FALSE)
```

```
 observation    x1    x2  y distance    yf position
           1  0.68  0.76  1     1.00  1.00       on
           2  1.80 -0.15  1     1.35  1.35   beyond
           3  1.38  0.66  1     1.50  1.50   beyond
           4  0.88  1.66  1     1.70  1.70   beyond
           5  0.62  2.34  1     1.90  1.90   beyond
           6  2.78 -0.04  1     2.20  2.20   beyond
           7  1.86  1.52  1     2.40  2.40   beyond
           8  0.50  0.40  1     0.64  0.64   inside
           9  0.10  0.20  1     0.20  0.20   inside
          10 -0.40  0.20  1     0.20 -0.20   inside
          11 -0.92 -0.44 -1     1.00  1.00       on
          12 -0.50 -1.50 -1     1.30  1.30   beyond
          13 -1.90 -0.05 -1     1.55  1.55   beyond
          14 -0.44 -2.33 -1     1.75  1.75   beyond
          15 -1.98 -0.61 -1     1.95  1.95   beyond
          16 -2.74  0.07 -1     2.15  2.15   beyond
          17 -1.90 -1.55 -1     2.45  2.45   beyond
          18 -0.50 -0.40 -1     0.64  0.64   inside
          19  0.00 -0.40 -1     0.24  0.24   inside
          20  0.40 -0.20 -1     0.20 -0.20   inside
```

Part b

```
cols <- ifelse(margin20$y == 1, "blue", "red")
inside <- margin20$position == "inside"

par(mar = c(4, 4, 1, 1))
plot(
  margin20$x1, margin20$x2, col = cols, pch = 16,
  xlim = c(-3.2, 3.2), ylim = c(-3, 3), asp = 1, xaxs = "i", yaxs = "i",
  xlab = expression(x[1]), ylab = expression(x[2])
)
# 0.8 x1 + 0.6 x2 = c  <=>  x2 = c / 0.6 - (4/3) x1
```

```r
abline(a = 0, b = -4 / 3, lwd = 2)
abline(a = 1 / 0.6, b = -4 / 3, lwd = 2, lty = 2)
abline(a = -1 / 0.6, b = -4 / 3, lwd = 2, lty = 2)
points(margin20$x1[inside], margin20$x2[inside], cex = 2, lwd = 1.5)
text(margin20$x1, margin20$x2, margin20$observation, pos = 3, cex = 0.6)
legend(
  "bottomleft", bty = "n", cex = 0.8,
  legend = c("Class 1", "Class -1", "Inside margin", "Boundary", "Margin lines"),
  col = c("blue", "red", "black", "black", "black"),
  pch = c(16, 16, 1, NA, NA), lty = c(NA, NA, NA, 1, 2), lwd = c(NA, NA, 1.5, 2, 2)
)
```

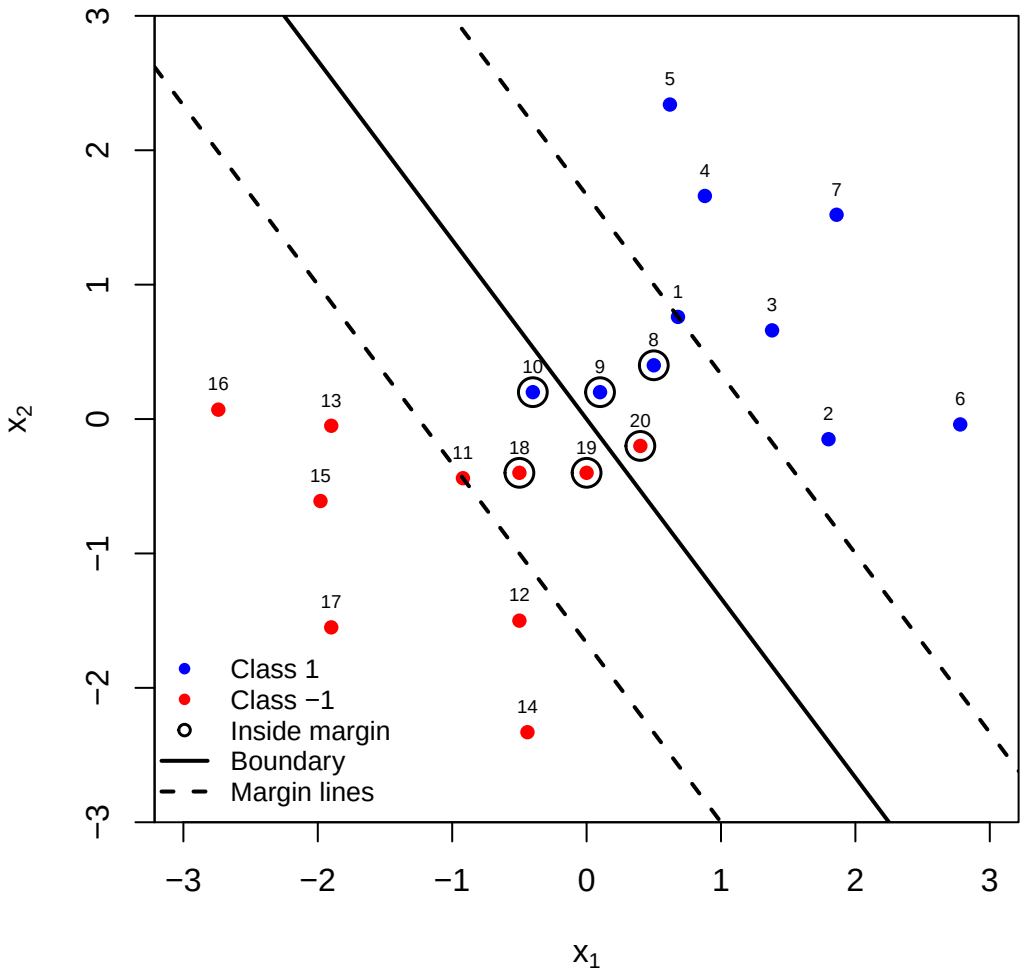

Figure 1: Observations with the boundary $f(x) = 0$ (solid) and margin lines $f(x) = \pm 1$ (dashed). Circled points are inside the margin.

Part c

```r
print(table(margin20$position))
```

```
beyond inside     on
```

```
    12       6       2
```

```
print(margin20$observation[margin20$yf < 0])
```

```
[1] 10 20
```

6 observations are inside the margin (8, 9, 10, 18, 19, 20), 2 are on it (1, 11), and 12 are beyond it. Observations 10 and 20 are misclassified since $yf(x) = -0.2 < 0$ places them on the wrong side of the boundary.

Question 2 Answers

Part a

The signed distance to the boundary is $f(\mathbf{x})/\|\beta\|_2$, with $\|\beta\|_2 = 5$. Projecting onto $\beta/\|\beta\|_2$ gives $\mathbf{x}^\mathsf{T}\beta/5$, and the boundary $f(\mathbf{x}) = 0$ sits at $-\beta_0/5 = 0.2$.

```
margin4d <- read.csv("data/margin-geometry-4d.csv")
X <- as.matrix(margin4d[, c("x1", "x2", "x3", "x4")])
y <- margin4d$y
beta0 <- -1
beta <- c(1, 2, -2, 4)
beta_norm <- sqrt(sum(beta^2))

projection <- as.vector(X %*% beta) / beta_norm
boundary <- -beta0 / beta_norm
geometric_dist <- y * (projection - boundary)
M <- min(geometric_dist)
closest <- abs(geometric_dist - M) < 1e-8

set.seed(432)
par(mar = c(4, 6, 1, 1))
plot(
  projection, y + runif(length(y), -0.1, 0.1),
  col = ifelse(y == 1, "blue", "red"), pch = 16,
  ylim = c(-1.4, 1.4), yaxt = "n", ylab = "",
  xlab = "Projection onto beta direction"
)
axis(2, at = c(-1, 1), labels = c("Class -1", "Class 1"), las = 1)
abline(v = boundary, lwd = 2)
abline(v = boundary + c(-M, M), lty = 2, lwd = 2)
points(projection[closest], y[closest], cex = 2)
```

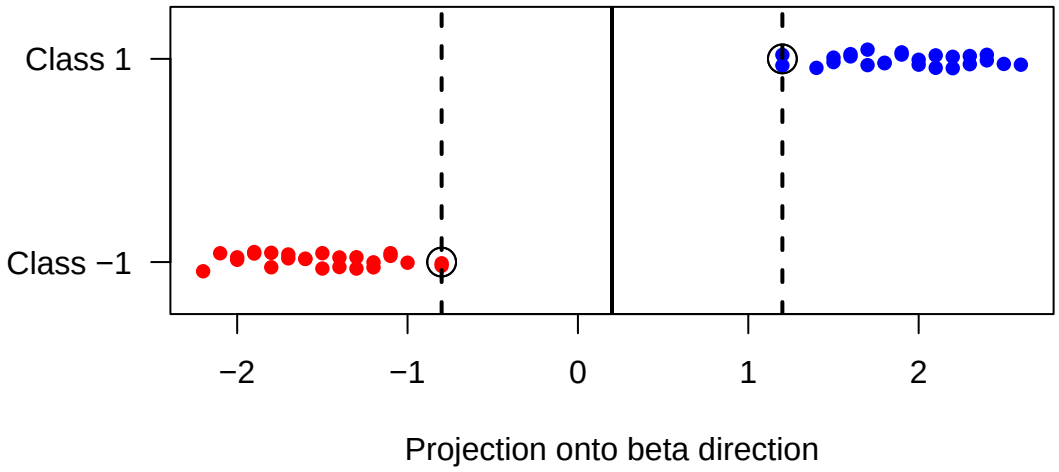

Figure 2: Projections onto the direction of β. Solid line: decision boundary. Dashed lines: closest observation in each class (circled).

```
print(c(boundary = boundary, M = M))
```

```
boundary        M
     0.2      1.0
```

```
print(margin4d$observation[closest])
```

```
[1]  4 15 33 36
```

The closest class -1 observations project to -0.8 and the closest class 1 observations to 1.2, each a distance 1 from the boundary at 0.2, so $M = 1$.

Part b

```
min_functional <- min(y * (beta0 + as.vector(X %*% beta)))
beta_svm <- beta / min_functional
beta0_svm <- beta0 / min_functional

print(min_functional)
```

```
[1] 5
```

```
print(c(beta0 = beta0_svm, beta = beta_svm))
```

```
beta0 beta1 beta2 beta3 beta4
 -0.2   0.2   0.4  -0.4   0.8
```

```
print(1 / sqrt(sum(beta_svm^2)))
```

```
[1] 1
```

The smallest functional margin is $\min_i y_i f(\mathbf{x}_i) = 5$, so dividing (β_0, β) by 5 gives the same boundary with

$$\beta = (0.2,\, 0.4,\, -0.4,\, 0.8)^{\mathsf{T}}, \qquad \beta_0 = -0.2.$$

Under this normalization $M = 1/\|\beta\|_2 = 1/1 = 1$, matching part a.

Question 3 Answers

Part a

I copied the skill definition from the assigned student's report into a `SKILL.md` file without making any changes. I then gave my agent the file path and asked it to follow the skill to analyze the SAheart dataset, using `chd` as the response and the other nine variables as predictors, coded as in Homework 7. I did not provide any additional preferences for the analysis or report.

The run took about 3.5 minutes and used around 45k tokens, including reading the original report. I did not need to steer or stop the agent. The skill produced `report.qmd` and `report.pdf`, and the results matched the author's report. The selected model was PCA+LDA with 7 principal components, a CV error of 0.246, and a test error of 0.293. The only issue was that the generated PDF was two pages instead of the required one page.

Part b

Overall, I thought the skill was clear and easy to follow. My agent was able to reproduce the same results as the original report without needing any extra guidance. The run took about 3.5 minutes and used around 45k tokens. I didn't need to steer or stop it. I liked that the skill included steps to prevent data leakage and keep the analysis consistent. However, I noticed a few things that could be improved.

1. One-page report requirement

The skill requires a one-page report, but mine ended up being two pages. I tried following the instructions to shorten the text and figure, but that wasn't enough. The author's report used additional formatting instructions that weren't included in the skill. I would recommend adding specific font sizes, margins, and code formatting so the one-page requirement can actually be met.

2. Model selection rules

The skill says to prefer simpler models when CV errors are within 0.005, but it doesn't clearly explain how to compare PCA models with regular LDA or QDA. I also noticed that 0.005 is pretty small compared to the standard errors of around 0.02. I tested 20 different seeds and the selected model changed several times. I would recommend defining the model complexity ranking more clearly and considering a one-standard-error rule to make model selection more consistent.

3. Classification performance

The selected model had a test error of 0.293, which was better than the baseline of 0.345. However, it only correctly identified 18 out of 40 heart disease cases, giving a sensitivity of 0.45. I think this is important to mention since the overall error doesn't tell the full story. I would recommend requiring sensitivity and specificity in the report so readers can better understand how the model performs.

Part c

https://github.com/PrerithArunkumar/stat432-fall2026/issues/3